

SQL Graduate Assessment 1

Question 1

Write a query in SQL to retrieve all rows and columns from the employee table in the Adventureworks database. Sort the result set in ascending order on jobtitle.

Answer:

```
SELECT *  
FROM HumanResources.Employee ORDER BY JobTitle
```

BusinessEntityID	NationalIDNumber	LoginID	OrganizationNode	OrganizationLevel	JobTitle	BirthDate	MaritalStatus	Gender	HireDate	SalariedFlag	VacationHours	SickLeaveHours
245	363910111	adventure-works\barbara1	0x5B20B0	3	Accountant	1976-01-04	M	F	2009-02-18	1	58	49
248	480951955	adventure-works\mike0	0x5B380	3	Accountant	1979-07-01	S	M	2009-03-08	1	59	49
241	30845	adventure-works\dauid6	0x5DA0	2	Accounts Manager	1983-07-08	M	M	2009-01-30	1	57	48
246	663843431	adventure-works\dragan0	0x5B180	3	Accounts Payable Specialist	1977-02-14	M	M	2009-02-11	0	63	51
247	519756660	adventure-works\janet0	0x5B280	3	Accounts Payable Specialist	1979-03-09	M	F	2009-03-01	0	64	52
242	363923697	adventure-works\deborah0	0x5B48	3	Accounts Receivable Specialist	1976-03-06	M	F	2008-12-18	0	60	50
243	60517918	adventure-works\candy0	0x5DA0	3	Accounts Receivable Specialist	1976-02-23	S	F	2009-01-06	0	61	50
244	931190412	adventure-works\byran1	0x5DAF	3	Accounts Receivable Specialist	1984-09-20	S	M	2009-01-24	0	62	51
267	58317344	adventure-works\karen1	0x5DA0	2	Application Specialist	1978-05-19	S	F	2009-02-16	1	74	57
268	314747499	adventure-works\ramesh0	0x5DE0	2	Application Specialist	1988-03-13	S	M	2009-02-03	1	73	56
269	671089528	adventure-works\dan0	0x5E10	2	Application Specialist	1987-05-26	M	M	2009-01-11	1	72	56
272	525932996	adventure-works\janaina0	0x5E70	2	Application Specialist	1985-01-30	M	F	2008-12-23	1	71	55
262	231203233	adventure-works\dauid5	0x5E10	2	Assistant to the Chief Financial Officer	1964-06-21	S	M	2009-01-12	0	56	48
239	619308550	adventure-works\mindy0	0x57080	3	Benefits Specialist	1984-11-20	M	F	2008-12-25	0	51	45
251	603686790	adventure-works\mike0	0x5EB58	4	Buyer	1984-08-17	S	M	2009-02-10	0	59	49

Question 2

Write a query in SQL to retrieve all rows and columns from the employee table using table aliasing in the Adventureworks database. Sort the output in ascending order on lastname.

Answer:

Select

```
e.BusinessEntityID,e.PersonType,e.NameStyle,e.Title,e.FirstName,e.MiddleName,e.LastName,e.  
Suffix,e.EmailPromotion,e.AdditionalContactInfo,e.Demographics,e.rowguid,e.ModifiedDate  
FROM Person.Person AS e Order by LastName
```

BusinessEntityID	PersonType	NameStyle	Title	FirstName	MiddleName	LastName	Suffix	EmailPromotion	AdditionalContactInfo	Demographics	rowguid
285	SP	0	Mr.	Syed	E	Abbas	NULL	0	NULL	IndividualSurvey	FF284881-01C2-4C...
293	SC	0	Ms.	Catherine	R	Abel	NULL	1	AdditionalContactInfo	IndividualSurvey	D54E0562-C226-4C...
295	SC	0	Ms.	Kim	NULL	Abercrombie	NULL	0	AdditionalContactInfo	IndividualSurvey	F7C8D848-0B44-47...
2170	GC	0	NULL	Kim	NULL	Abercrombie	NULL	2	NULL	IndividualSurvey	24F01854-7A67-4B...
38	EM	0	NULL	Kim	B	Abercrombie	NULL	2	NULL	IndividualSurvey	5A216383-2F4D-4F...
211	EM	0	NULL	Hazem	E	Abolrous	NULL	0	NULL	IndividualSurvey	C2637051-25A5-4A...
2357	GC	0	NULL	Sam	NULL	Abolrous	NULL	1	NULL	IndividualSurvey	7888F41E-ED14-4E...
297	SC	0	Sr.	Humberto	NULL	Acevedo	NULL	2	AdditionalContactInfo	IndividualSurvey	5A41D336-84CF-44...
291	SC	0	Mr.	Gustavo	NULL	Achong	NULL	2	AdditionalContactInfo	IndividualSurvey	D4C132D3-FCB5-42...
299	SC	0	Sra.	Pilar	NULL	Ackerman	NULL	0	AdditionalContactInfo	IndividualSurvey	0F1F8B48-2323-43...
121	EM	0	NULL	Pilar	G	Ackerman	NULL	0	NULL	IndividualSurvey	81F50324-C0B5-4E...
16867	IN	0	NULL	Aaron	B	Adams	NULL	2	NULL	IndividualSurvey	846210F5-A3C8-46...
16901	IN	0	NULL	Adam	NULL	Adams	NULL	1	NULL	IndividualSurvey	16C926A9-8BAE-4C...
16724	IN	0	NULL	Alex	C	Adams	NULL	0	NULL	IndividualSurvey	A99994C0-C0EE-42...
10263	IN	0	NULL	Alexandra	J	Adams	NULL	0	NULL	IndividualSurvey	90A13D3C-0C9C-44...

Question 3

Write a query in SQL to return all rows and a subset of the columns (FirstName, LastName, businessentityid) from the person table in the AdventureWorks database. The third column heading is renamed to Employee_id. Arranged the output in ascending order by lastname.

Answer:

```
SELECT FirstName, LastName, BusinessEntityID AS employee_id
FROM Person.Person ORDER BY LastName
```



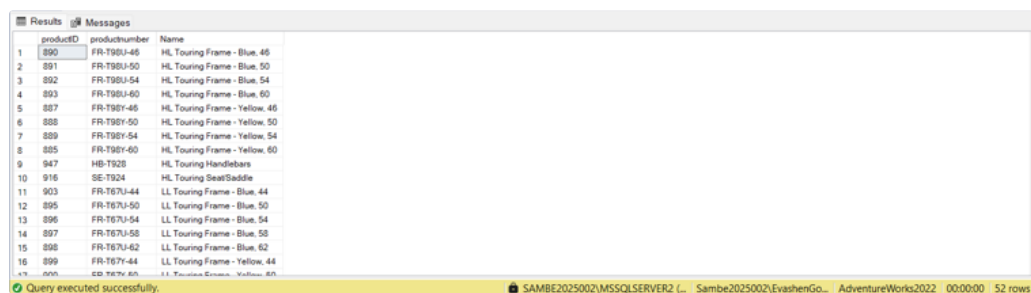
	FirstName	LastName	employee_id
1	Syed	Abbas	285
2	Catherine	Abel	293
3	Kim	Abercrombie	295
4	Kim	Abercrombie	2170
5	Kim	Abercrombie	38
6	Hazem	Abolrous	211
7	Sam	Abolrous	2357
8	Humberto	Akocido	297
9	Quintus	Achong	291
10	Pilar	Ackerman	299
11	Pilar	Ackerman	121
12	Aaron	Adams	16867
13	Adam	Adams	16901
14	Alex	Adams	16724
15	Alexandra	Adams	10263
16	Allison	Adams	10312
17	Amanda	Adams	10374

Question 4

Write a query in SQL to return only the rows for product that have a sellstartdate that is not NULL and a productline of 'T'. Return productid, productnumber, and name. Arranged the output in ascending order on name.

Answer:

```
SELECT productID, productnumber, Name FROM Production.Product
WHERE SellStartDate IS NOT NULL AND ProductLine = 'T'
ORDER BY Name ASC
```



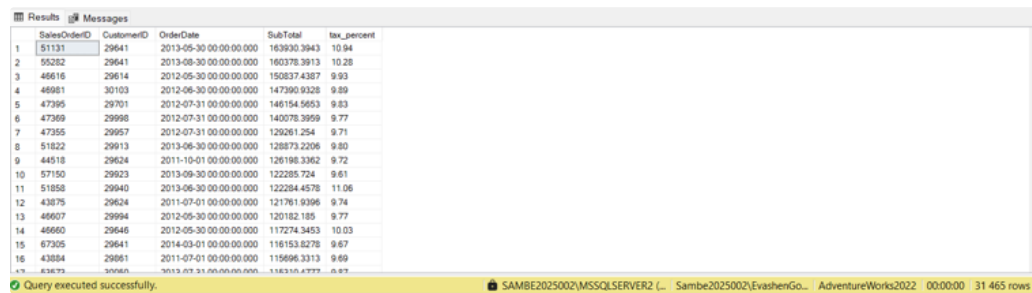
	productID	productnumber	Name
1	890	FR-T98U-46	HL Touring Frame - Blue, 46
2	891	FR-T98U-50	HL Touring Frame - Blue, 50
3	892	FR-T98U-54	HL Touring Frame - Blue, 54
4	893	FR-T98U-60	HL Touring Frame - Blue, 60
5	897	FR-T98Y-46	HL Touring Frame - Yellow, 46
6	898	FR-T98Y-50	HL Touring Frame - Yellow, 50
7	899	FR-T98Y-54	HL Touring Frame - Yellow, 54
8	895	FR-T98Y-60	HL Touring Frame - Yellow, 60
9	947	HB-T928	HL Touring Handlebars
10	916	SE-T924	HL Touring Seat/Saddle
11	903	FR-T67U-44	LL Touring Frame - Blue, 44
12	895	FR-T67U-50	LL Touring Frame - Blue, 50
13	896	FR-T67U-54	LL Touring Frame - Blue, 54
14	897	FR-T67U-58	LL Touring Frame - Blue, 58
15	898	FR-T67U-62	LL Touring Frame - Blue, 62
16	899	FR-T67Y-44	LL Touring Frame - Yellow, 44
17	900	ED-T67Y-60	LL Touring Frame - Yellow, 60

Question 5

Write a query in SQL to return all rows from the salesorderheader table in Adventureworks database and calculate the percentage of tax on the subtotal have decided. Return salesorderid, customerid, orderdate, subtotal, percentage of tax column. Arranged the result set in ascending order on subtotal.

Answer:

```
SELECT SalesOrderID, CustomerID, OrderDate, SubTotal, TaxAmt/SubTotal * 100 AS tax_percent
from Sales.SalesOrderHeader
ORDER BY SubTotal DESC
```



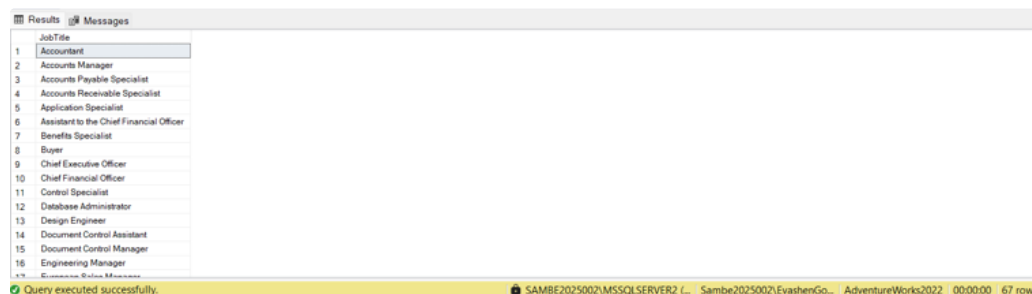
	SalesOrderID	CustomerID	OrderDate	SubTotal	tax_percent
1	51131	29641	2013-05-30 00:00:00.000	163930.3943	10.94
2	55352	29641	2013-06-30 00:00:00.000	160378.3913	10.26
3	46616	29614	2012-06-30 00:00:00.000	150337.4387	9.93
4	46681	30103	2012-06-30 00:00:00.000	147390.9328	9.89
5	47395	29701	2012-07-31 00:00:00.000	146154.5653	9.83
6	47369	29998	2012-07-31 00:00:00.000	140078.3959	9.77
7	47355	29957	2012-07-31 00:00:00.000	129261.254	9.71
8	51822	29913	2013-06-30 00:00:00.000	128873.2206	9.80
9	44518	29624	2011-10-01 00:00:00.000	126186.3362	9.72
10	57150	29923	2013-06-30 00:00:00.000	122285.724	9.61
11	51858	29940	2013-06-30 00:00:00.000	122284.4578	11.06
12	43875	29624	2011-07-01 00:00:00.000	121761.9396	9.74
13	46607	29994	2012-05-30 00:00:00.000	120182.185	9.77
14	46660	29646	2012-05-30 00:00:00.000	117274.3453	10.03
15	67305	29641	2014-03-01 00:00:00.000	116153.8278	9.67
16	43854	29651	2011-07-01 00:00:00.000	115696.3313	9.69

Question 6

Write a query in SQL to create a list of unique jobtitles in the employee table in Adventureworks database. Return jobtitle column and arranged the resultset in ascending order.

Answer:

```
SELECT DISTINCT JobTitle FROM HumanResources.Employee
ORDER BY JobTitle
```



	JobTitle
1	Accountant
2	Accounts Manager
3	Accounts Payable Specialist
4	Accounts Receivable Specialist
5	Application Specialist
6	Assistant to the Chief Financial Officer
7	Benefits Specialist
8	Buyer
9	Chief Executive Officer
10	Chief Financial Officer
11	Control Specialist
12	Database Administrator
13	Design Engineer
14	Document Control Assistant
15	Document Control Manager
16	Engineering Manager

Question 7

Write a query in SQL to calculate the total freight paid by each customer. Return customerid and total freight. Sort the output in ascending order on customerid.

Answer:

```
SELECT CustomerID, SUM(Freight) AS total_freight FROM Sales.SalesOrderHeader  
GROUP BY CustomerID  
ORDER BY CustomerID
```



The screenshot shows the results of a SQL query in a window titled 'Results Messages'. The results pane displays a table with two columns: 'CustomerID' and 'total_freight'. The data is sorted by CustomerID in ascending order. The status bar at the bottom indicates 'Query executed successfully.' and '19 119 rows'.

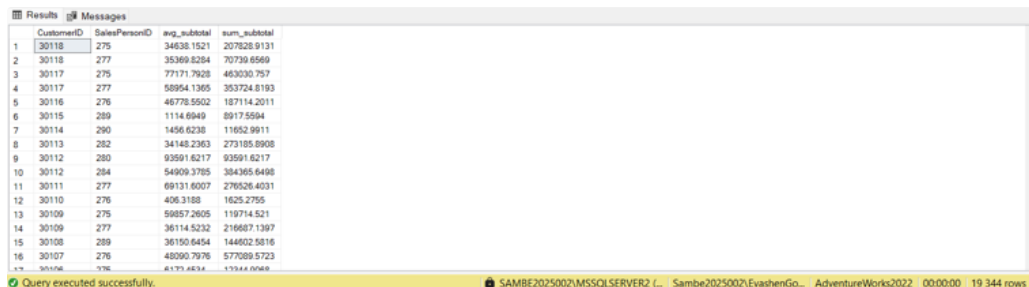
CustomerID	total_freight
11000	206.2249
11001	159.5971
11002	202.8511
11003	203.4823
11004	204.9003
11005	203.0334
11006	202.9759
11007	205.2751
11008	202.6578
11009	202.2833
11010	202.2011
11011	203.3261
11012	2.0315
11013	2.849
11014	3.4613
11015	62.5243

Question 8

Write a query in SQL to find the average and the sum of the subtotal for every customer. Return customerid, average and sum of the subtotal. Grouped the result on customerid and salespersonid. Sort the result on customerid column in descending order.

Answer:

```
SELECT CustomerID,SalesPersonID,AVG(SubTotal) AS avg_subtotal, SUM(SubTotal) AS  
sum_subtotal FROM Sales.SalesOrderHeader  
GROUP BY CustomerID,SalesPersonID  
ORDER BY CustomerID DESC
```



The screenshot shows the results of a SQL query in a window titled 'Results Messages'. The results pane displays a table with four columns: 'CustomerID', 'SalesPersonID', 'avg_subtotal', and 'sum_subtotal'. The data is sorted by CustomerID in descending order. The status bar at the bottom indicates 'Query executed successfully.' and '19 344 rows'.

CustomerID	SalesPersonID	avg_subtotal	sum_subtotal
30118	275	34638.1521	207628.9131
30118	277	35369.8284	70739.6569
30117	275	77171.7928	463030.757
30117	277	58954.1365	353724.8193
30116	276	46778.5502	187114.2011
30115	289	1114.6949	8917.5594
30114	290	1456.8238	11652.9911
30113	282	34148.2363	273185.8908
30112	280	93591.6217	93591.6217
30112	284	54909.3785	384365.6498
30111	277	69131.6007	278526.4031
30110	276	406.3188	1625.2755
30109	275	59857.2605	119714.521
30109	277	36114.5232	216687.1397
30108	289	36150.6454	144052.5816
30107	276	48090.7976	577089.5723
30106	276	61373.4834	170344.1048

Question 9

Write a query in SQL to retrieve total quantity of each productid which are in shelf of 'A' or 'C' or 'H'. Filter the results for sum quantity is more than 500. Return productid and sum of the quantity. Sort the results according to the productid in ascending order.

Answer:

```
SELECT ProductID,SUM(Quantity) AS total_quantity FROM Production.ProductInventory
WHERE Shelf IN('A','C','H')
GROUP BY ProductID
HAVING SUM(Quantity) > 500
ORDER BY ProductID
```



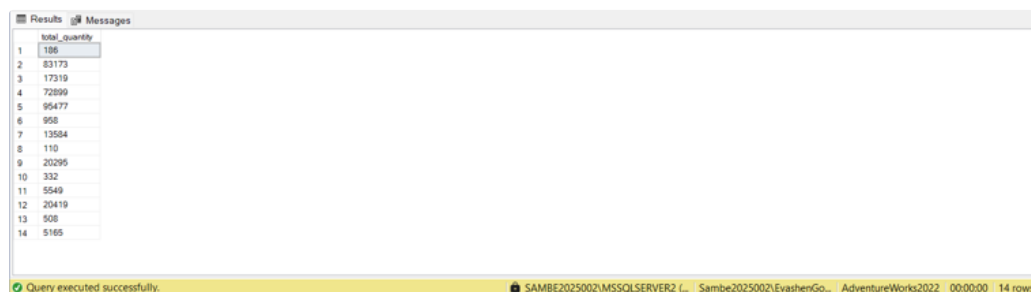
	ProductID	total_quantity
1	1	761
2	2	791
3	3	909
4	4	900
5	316	532
6	317	593
7	319	797
8	320	1136
9	321	1750
10	322	1684
11	323	1684
12	324	1629
13	325	1210
14	326	1097
15	328	1044
16	329	1025

Question 10

Write a query in SQL to find the total quantity for a group of locationid multiplied by 10.

Answer:

```
Select SUM(Quantity) AS total_quantity FROM Production.ProductInventory
GROUP BY LocationID
```



	LocationID	total_quantity
1	1	186
2	2	83173
3	3	17319
4	4	72899
5	5	95477
6	6	958
7	7	13584
8	8	110
9	9	20295
10	10	332
11	11	5549
12	12	20419
13	13	508
14	14	5165

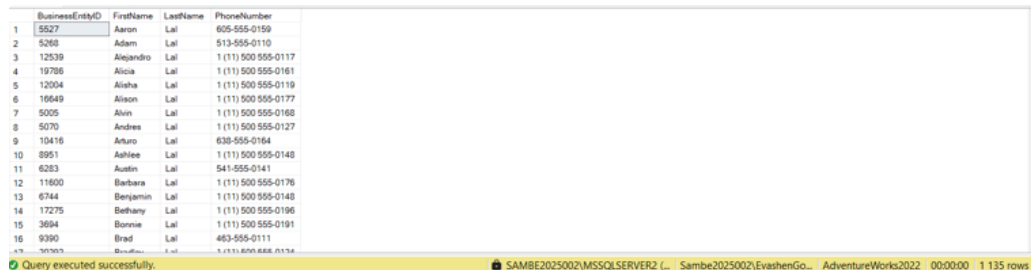
Question 11

Write a query in SQL to find the persons whose last name starts with letter 'L'. Return BusinessEntityID, FirstName, LastName, and PhoneNumber. Sort the result on lastname and

firstname.

Answer:

```
SELECT p.BusinessEntityID,p.FirstName,p.LastName,pp.PhoneNumber From Person.Person as  
p JOIN Person.PersonPhone as pp  
ON p.BusinessEntityID = pp.BusinessEntityID  
WHERE LastName LIKE 'L%'  
ORDER BY LastName,FirstName
```



	BusinessEntityID	FirstName	LastName	PhoneNumber
1	5527	Aaron	Lai	605-555-0159
2	5268	Adam	Lai	513-555-0110
3	12539	Alejandro	Lai	1 (11) 500 555-0117
4	19786	Alicia	Lai	1 (11) 500 555-0161
5	12004	Alisha	Lai	1 (11) 500 555-0119
6	16649	Alison	Lai	1 (11) 500 555-0177
7	5005	Alvin	Lai	1 (11) 500 555-0168
8	5070	Andres	Lai	1 (11) 500 555-0127
9	10416	Arturo	Lai	638-555-0164
10	8951	Ashlee	Lai	1 (11) 500 555-0148
11	6283	Austin	Lai	541-555-0141
12	11600	Barbara	Lai	1 (11) 500 555-0176
13	6744	Benjamin	Lai	1 (11) 500 555-0148
14	17275	Bethany	Lai	1 (11) 500 555-0196
15	3694	Bonnie	Lai	1 (11) 500 555-0191
16	9390	Brad	Lai	463-555-0111

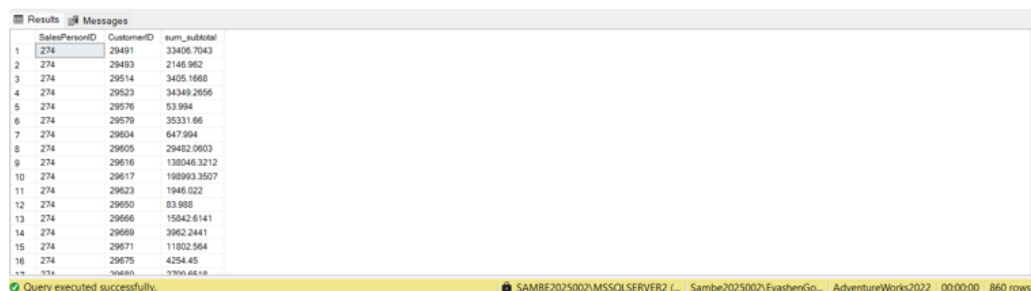
Query executed successfully. SAMBE2025002\MSSQLSERVER2 (S... Sambe2025002\EvashenGo... AdventureWorks2022 00:00:00 1 135 rows

Question 12

Write a query in SQL to find the sum of subtotal column. Group the sum on distinct salespersonid and customerid. Rolls up the results into subtotal and running total. Return salespersonid, customerid and sum of subtotal column i.e. sum_subtotal.

Answer:

```
SELECT DISTINCT(SalesPersonID), CustomerID, SUM(SubTotal) AS sum_subtotal  
FROM Sales.SalesOrderHeader  
WHERE SalesPersonID IS NOT NULL  
GROUP BY SalesPersonID, CustomerID  
ORDER BY SalesPersonID ASC
```



	SalesPersonID	CustomerID	sum_subtotal
1	274	29491	33406.7043
2	274	29493	2146.962
3	274	29514	3405.1668
4	274	29523	34349.2656
5	274	29576	53.994
6	274	29579	35331.66
7	274	29604	647.994
8	274	29605	29482.0603
9	274	29616	138046.3212
10	274	29617	190993.3507
11	274	29623	1946.022
12	274	29650	83.988
13	274	29666	15042.6141
14	274	29669	3962.2441
15	274	29671	11802.564
16	274	29675	4254.45

Query executed successfully. SAMBE2025002\MSSQLSERVER2 (S... Sambe2025002\EvashenGo... AdventureWorks2022 00:00:00 860 rows

Question 13

Write a query in SQL to find the sum of the quantity of all combination of group of distinct locationid and shelf column. Return locationid, shelf and sum of quantity as TotalQuantity.

Answer:

```
SELECT DISTINCT LocationID,Shelf, SUM(Quantity) AS totalquantity
FROM Production.ProductInventory
GROUP BY LocationID,Shelf
ORDER BY SUM(Quantity) DESC
```



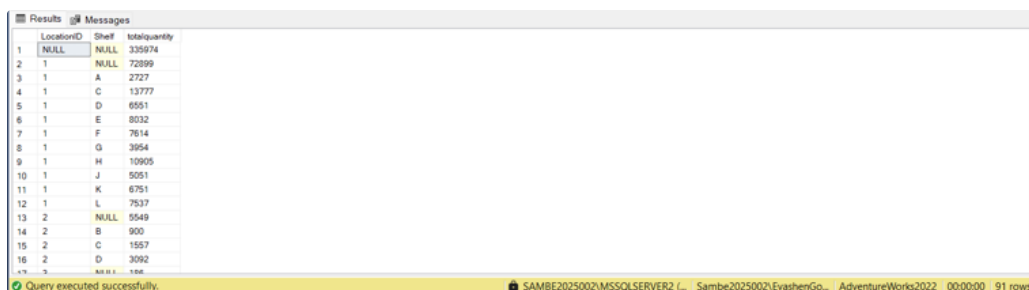
LocationID	Shelf	totalquantity
50	1	18190
50	E	17775
7	NA	17319
6	R	16977
1	C	13777
1	H	10905
50	A	10329
6	H	9150
50	K	8881
60	NA	8804
1	E	8032
50	T	7805
1	F	7614
1	L	7537
1	K	6751
60	G	6664

Question 14

Write a query in SQL to find the sum of the quantity with subtotal for each locationid. Group the results for all combination of distinct locationid and shelf column. Rolls up the results into subtotal and running total. Return locationid, shelf and sum of quantity as TotalQuantity.

Answer:

```
SELECT DISTINCT LocationID,Shelf, SUM(Quantity) AS totalquantity
FROM Production.ProductInventory
GROUP BY ROLLUP(LocationID,Shelf)
```



LocationID	Shelf	totalquantity
NULL	NULL	335974
1	NULL	72899
1	A	2727
1	C	13777
1	D	6551
1	E	8032
1	F	7614
1	G	3954
1	H	10905
1	J	5051
1	K	6751
1	L	7537
2	NULL	5549
2	B	900
2	C	1557
2	D	3092
3	NA	17319

Question 15

Write a query in SQL to find the total quantity for each locationid and calculate the grandtotal for all locations. Return locationid and total quantity. Group the results on locationid.

Answer:

```
SELECT DISTINCT LocationID, SUM(Quantity) AS totalquantity
FROM Production.ProductInventory
GROUP BY ROLLUP(LocationID)
```



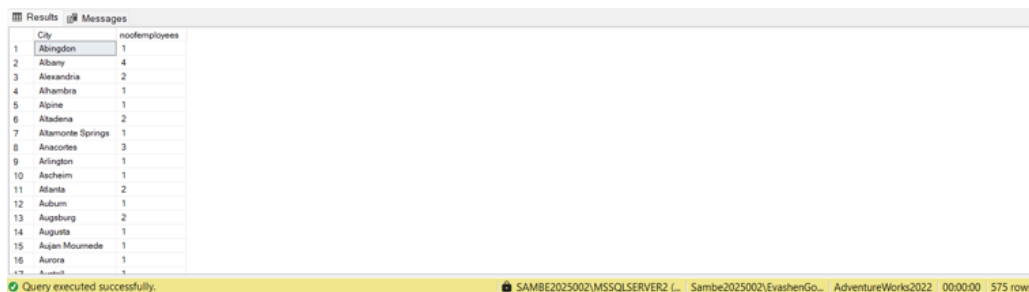
LocationID	totalquantity
NULL	135074
1	9549
2	186
3	110
4	20295
5	83173
6	17319
7	13584
10	5165
20	958
30	505
40	332
45	95477
50	20419
60	

Question 16

Write a query in SQL to retrieve the number of employees for each City. Return city and number of employees. Sort the result in ascending order on city.

Answer:

```
SELECT City,COUNT(AddressID) AS noofemployees
FROM Person.Address
GROUP BY City
ORDER BY City
```



City	noofemployees
Abingdon	1
Albany	4
Alexandria	2
Alhambra	1
Alpine	1
Altadena	2
Altamonte Springs	1
Anacortes	3
Arlington	1
Ascheim	1
Atlanta	2
Auburn	1
Augsburg	2
Augusta	1
Aujan Moumède	1
Aurora	1

Question 17

Write a query in SQL to retrieve the total sales for each year. Return the year part of order date and total due amount. Sort the result in ascending order on year part of order date.

Answer:

```
SELECT YEAR(OrderDate) AS Year,SUM(TotalDue) AS totaldueorder
FROM Sales.SalesOrderHeader
GROUP BY YEAR(OrderDate)
ORDER BY YEAR(OrderDate) ASC
```



The screenshot shows a SQL Server query results window with the following data:

	Year	totaldueorder
1	2011	14155699.525
2	2012	37675700.312
3	2013	48965887.9632
4	2014	22419498.3157

Query executed successfully. SAMBE2025002\MSSQLSERVER2 (S) Sambe2025002\EvashenGo... AdventureWorks2022 00:00:00 4 rows

Question 18

Write a query in SQL to retrieve the total sales for each year. Filter the result set for those orders where order year is on or before 2016. Return the year part of orderdate and total due amount. Sort the result in ascending order on year part of order date.

Answer:

```
SELECT YEAR(OrderDate) AS yearoforderdate,SUM(TotalDue) AS totaldueorder
FROM Sales.SalesOrderHeader
GROUP BY YEAR(OrderDate)
HAVING YEAR(OrderDate) <= 2016
ORDER BY YEAR(OrderDate) ASC
```



The screenshot shows a SQL Server query results window with the following data:

	yearoforderdate	totaldueorder
1	2011	14155699.525
2	2012	37675700.312
3	2013	48965887.9632
4	2014	22419498.3157

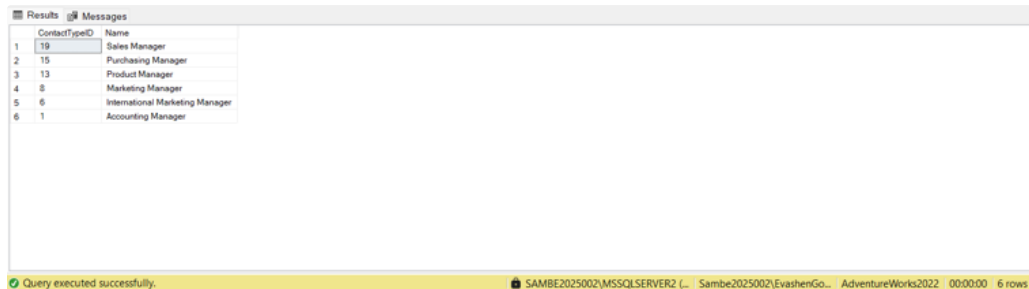
Query executed successfully. SAMBE2025002\MSSQLSERVER2 (S) Sambe2025002\EvashenGo... AdventureWorks2022 00:00:00 4 rows

Question 19

Write a query in SQL to find the contacts who are designated as a manager in various departments. Returns ContactTypeID, name. Sort the result set in descending order.

Answer:

```
SELECT ContactTypeID,Name
FROM Person.ContactType
WHERE Name LIKE '%Manager%'
ORDER BY ContactTypeID DESC
```



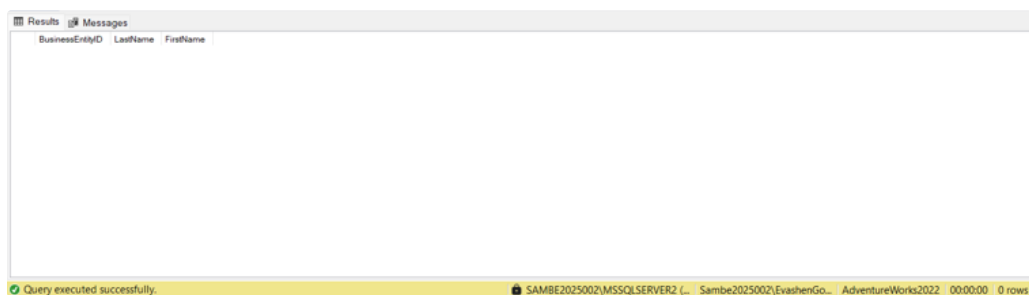
	ContactTypeID	Name
1	19	Sales Manager
2	15	Purchasing Manager
3	13	Product Manager
4	8	Marketing Manager
5	6	International Marketing Manager
6	1	Accounting Manager

Question 20

From the following tables write a query in SQL to make a list of contacts who are designated as 'Purchasing Manager'. Return BusinessEntityID, LastName, and FirstName columns. Sort the result set in ascending order of LastName, and FirstName.

Answer:

```
SELECT p.BusinessEntityID,p.LastName,p.FirstName
FROM Person.Person AS p
JOIN Person.BusinessEntityContact AS b
ON p.BusinessEntityID = b.BusinessEntityID
JOIN Person.ContactType c
ON b.ContactTypeID = c.ContactTypeID
WHERE c.Name = 'Purchasing Manager'
ORDER BY p.LastName,p.FirstName
```



BusinessEntityID	LastName	FirstName
------------------	----------	-----------

The data is not linked.

Question 21

Write a query in SQL to retrieve the salesperson for each PostalCode who belongs to a territory and SalesYTD is not zero. Return row numbers of each group of PostalCode, last name, salesytd, postalcode column. Sort the salesytd of each postalcode group in descending order. Sorts the postalcode in ascending order.

Answer:

```
SELECT ROW_NUMBER() OVER(PARTITION BY a.PostalCode ORDER BY s.SalesYTD DESC) AS RowNum,  
P.LastName, s.SalesYTD, a.PostalCode  
FROM Sales.SalesPerson AS s  
JOIN Person.Person AS p ON s.BusinessEntityID = p.BusinessEntityID  
JOIN Person.BusinessEntityAddress AS b ON p.BusinessEntityID = b.BusinessEntityID  
JOIN Person.Address AS a ON b.AddressID = a.AddressID  
ORDER BY s.SalesYTD DESC, a.PostalCode ASC
```



RowNum	LastName	SalesYTD	PostalCode
1	Mitchell	4251368.5497	84407
2	Pak	4116871.2277	8A5 3H6
3	Blythe	3763178.1787	48226
4	Carson	3189418.3662	55802
5	Vietkey Chudakul	3121816.3262	33080
6	Saraisa	2604540.7172	K4B 117
7	Ito	2458535.6169	94109
8	Reiter	2315185.6111	38103
9	Valdez	1827066.7118	14111
10	Mensa-Annan	1576562.1966	02139
11	Campbell	1573012.9383	98004
12	Vargas	1453719.4953	T2P 208
13	Toufex	1421810.8242	3000
14	Annaman-Wolle	1352577.1325	97205
15	Jiang	559697.5639	98052
16	Alberts	519905.932	98055
17	Robinson	179676.4813	08011

Question 22

Write a query in SQL to count the number of contacts for combination of each type and name. Filter the output for those who have 100 or more contacts. Return ContactTypeID and ContactTypeName and BusinessEntityContact. Sort the result set in descending order on number of contacts.

Answer:

```
SELECT b.ContactTypeID,c.Name,COUNT(b.ContactTypeID) AS nocontacts  
FROM Person.BusinessEntityContact AS b  
JOIN Person.ContactType AS c  
ON b.ContactTypeID = c.ContactTypeID  
GROUP BY b.ContactTypeID,c.Name  
HAVING COUNT(b.ContactTypeID) >= 100  
ORDER BY COUNT(b.ContactTypeID) DESC
```

Results Messages			
ContactTypeID	Name	nocontacts	
11	Owner	266	
15	Purchasing Manager	245	
14	Purchasing Agent	242	

Query executed successfully. SAMBE2025002\MSSQLSERVER2 (S) Sambe2025002\EvashenGo... AdventureWorks2022 00:00:00 3 rows

Question 23

Write a query in SQL to retrieve the RateChangeDate, full name (first name, middle name and last name) and weekly salary (40 hours in a week) of employees. In the output the RateChangeDate should appears in date format. Sort the output in ascending order on NameInFull.

Answer:

```
SELECT FORMAT(e.RateChangeDate,'yyyy-MM-dd') AS fromdate , CONCAT(p.FirstName, ' ',
,p.MiddleName, ' ',LEFT(p.LastName,1)) AS nameinfull , (e.Rate * 40) AS salaryinweek
FROM HumanResources.EmployeePayHistory AS e
JOIN Person.Person AS p
ON e.BusinessEntityID = p.BusinessEntityID
ORDER BY nameinfull ASC
```

Results Messages			
fromdate	nameinfull	salaryinweek	
2008-12-17	Alexandro, E M	600.00	
2009-02-17	Alice, O C	440.00	
2009-02-08	Alice, O C	440.00	
2008-12-07	Alice, O C	440.00	
2012-04-16	Amy, E A	1924.04	
2009-02-02	Andreas, T B	423.076	
2009-03-06	Andreas, M C	400.00	
2009-02-22	Andreas, R H	1000.00	
2009-01-31	Andy, M R	380.00	
2009-01-20	Angela, W B	440.00	
2009-02-23	Arbail, T S	560.00	
2010-12-06	Annette, L H	510.00	
2008-12-17	Annik, O S	498.00	
2009-02-28	Arvind, B R	730.768	
2008-12-04	Ashwini, R S	1298.076	

Query executed successfully. SAMBE2025002\MSSQLSERVER2 (S) Sambe2025002\EvashenGo... AdventureWorks2022 00:00:00 316 rows

Question 24

Write a query in SQL to calculate and display the latest weekly salary of each employee. Return RateChangeDate, full name (first name, middle name and last name) and weekly salary (40 hours in a week) of employees Sort the output in ascending order on NameInFull.

Answer:

```
SELECT FORMAT(e.RateChangeDate,'yyyy-MM-dd') AS fromdate , CONCAT(p.FirstName, ' ',
,p.MiddleName, ' ',LEFT(p.LastName,1)) AS nameinfull , (e.Rate * 40) AS salaryinweek
```

```

FROM HumanResources.EmployeePayHistory AS e
JOIN Person.Person AS p
ON e.BusinessEntityID = p.BusinessEntityID
WHERE e.RateChangeDate = (
    SELECT MAX(e2.RateChangeDate)
    FROM HumanResources.EmployeePayHistory AS e2
    WHERE e.BusinessEntityID = e2.BusinessEntityID)
ORDER BY nameinfull ASC

```

	RateDate	NameInFull	Salary
1	2008-12-12	A. Scott, W	942.308
2	2009-02-13	Alan, J B	640.00
3	2008-12-06	Alejandro, E M	600.00
4	2009-02-08	Alex, M N	400.00
5	2009-12-07	Alice, O C	440.00
6	2012-04-16	Amy, E A	1904.04
7	2009-02-02	Andrew, T B	423.076
8	2009-03-06	Andrew, M C	400.00
9	2009-02-22	Andrew, R H	1000.00
10	2009-01-31	Andy, M R	380.00
11	2009-01-20	Angela, W B	440.00
12	2009-02-23	Anibal, T S	560.00
13	2010-12-06	Annette, L H	510.00
14	2009-12-17	Annik, O S	490.00
15	2009-02-28	Arnold, B R	730.768
16	2008-12-04	Ashwin, R S	1298.076
17	2009-03-18	Bashir, C M	1067.603

Question 25

Write a query in SQL to find the sum, average, count, minimum, and maximum order quantity for those orders whose id are 43659 and 43664. Return SalesOrderID, ProductID, OrderQty, sum, average, count, max, and min order quantity.

Answer:

```

SELECT SalesOrderID,ProductID,OrderQty,SUM(OrderQty) OVER() AS 'Total Quantity',
AVG(OrderQty) OVER() AS 'Avg Quantity', COUNT(OrderQty) OVER() AS 'No of Orders',
MIN(OrderQty) OVER() AS 'Min Quantity', MAX(OrderQty) OVER() AS 'Max Quantity'
FROM Sales.SalesOrderDetail
WHERE SalesOrderID = 43659 OR SalesOrderID = 43664

```

	SalesOrderID	ProductID	OrderQty	Total Quantity	Avg Quantity	No of Orders	Min Quantity	Max Quantity
1	43659	776	1	40	2	20	1	6
2	43659	777	3	40	2	20	1	6
3	43659	778	1	40	2	20	1	6
4	43659	771	1	40	2	20	1	6
5	43659	772	1	40	2	20	1	6
6	43659	773	2	40	2	20	1	6
7	43659	774	1	40	2	20	1	6
8	43659	714	3	40	2	20	1	6
9	43659	716	1	40	2	20	1	6
10	43659	709	6	40	2	20	1	6
11	43659	712	2	40	2	20	1	6
12	43659	711	4	40	2	20	1	6
13	43664	772	1	40	2	20	1	6
14	43664	775	4	40	2	20	1	6
15	43664	714	1	40	2	20	1	6
16	43664	716	1	40	2	20	1	6
17	43664	777	3	40	2	20	1	6